

Figure 4: Adaptive Therapy via the ABC and VEGF Models

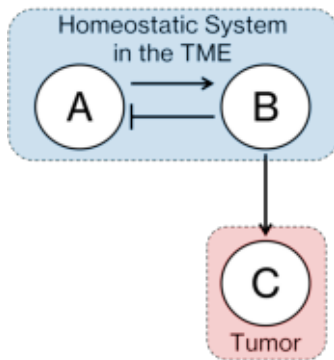
A

ABC Model

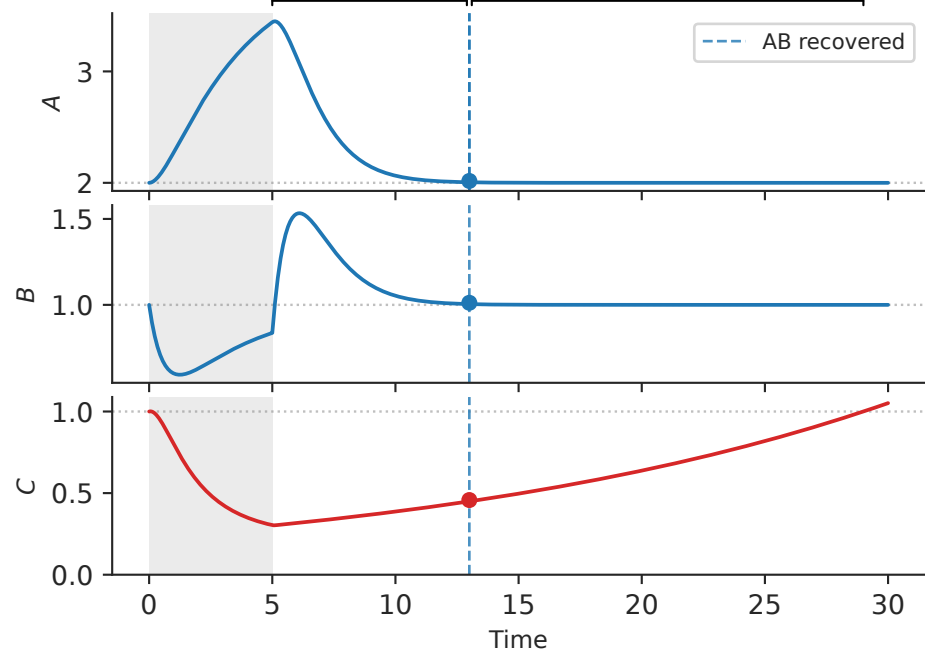
$$\frac{dA}{dt} = a_1(\mu - B)$$

$$\frac{dB}{dt} = b_1A - b_2B$$

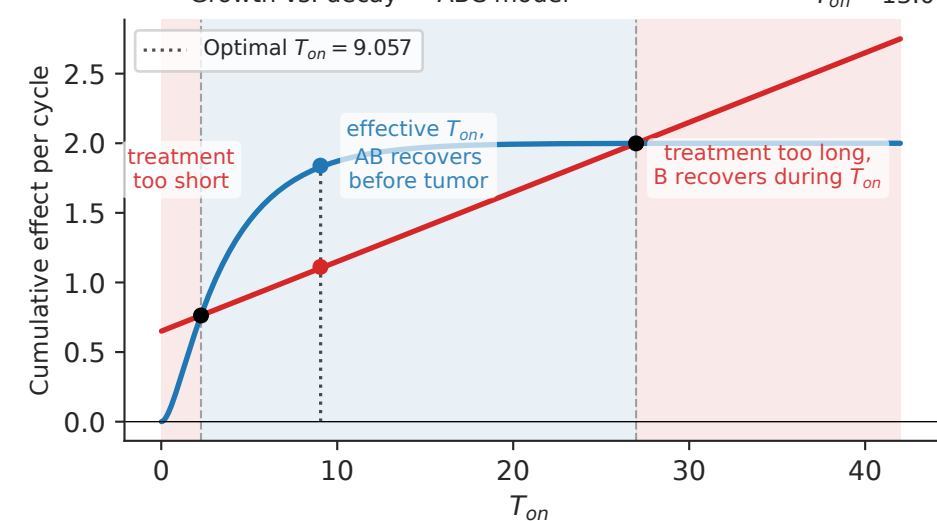
$$\frac{dC}{dt} = C(c_1 - c_2(\mu - B)^+)$$



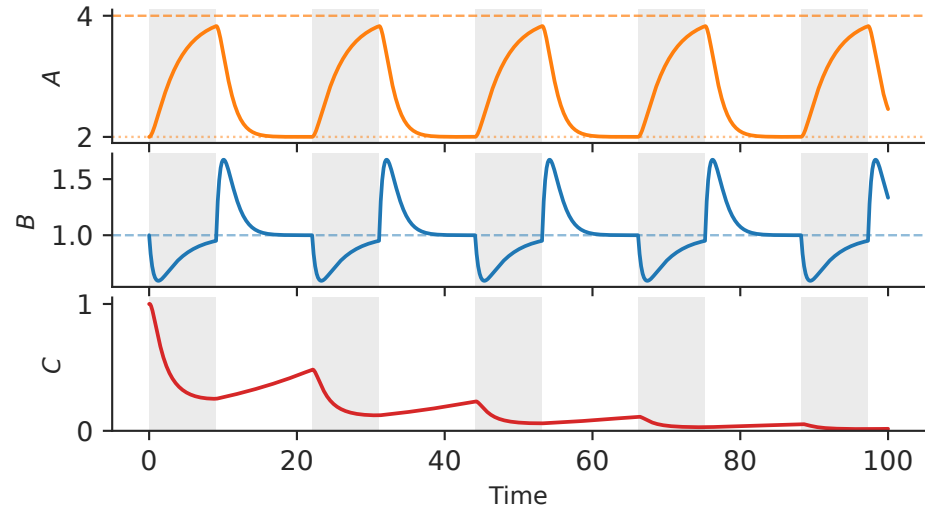
Timescale separation: the homeostatic system recovers before the tumor
 fast recovery of AB system Tumor has not recovered yet, can apply another treatment cycle

B**C**

Growth vs. decay — ABC model

 $T_{off} = 13.0$ 

Adaptive therapy for the optimal treatment time found
 $(T_{on} = 9.1, T_{off} = 13.0)$

D**E**

Angiogenesis Model

Same principles as ABC model, but with redundancy and feedback

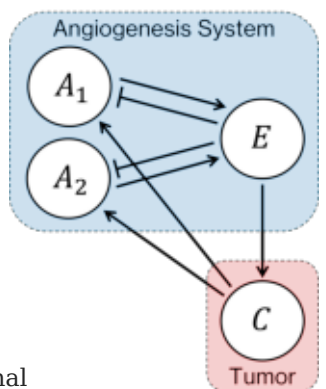
$$\frac{dA_1}{dt} = k_1(\mu C - E)$$

$$\frac{dA_2}{dt} = k_2(\mu C - E)$$

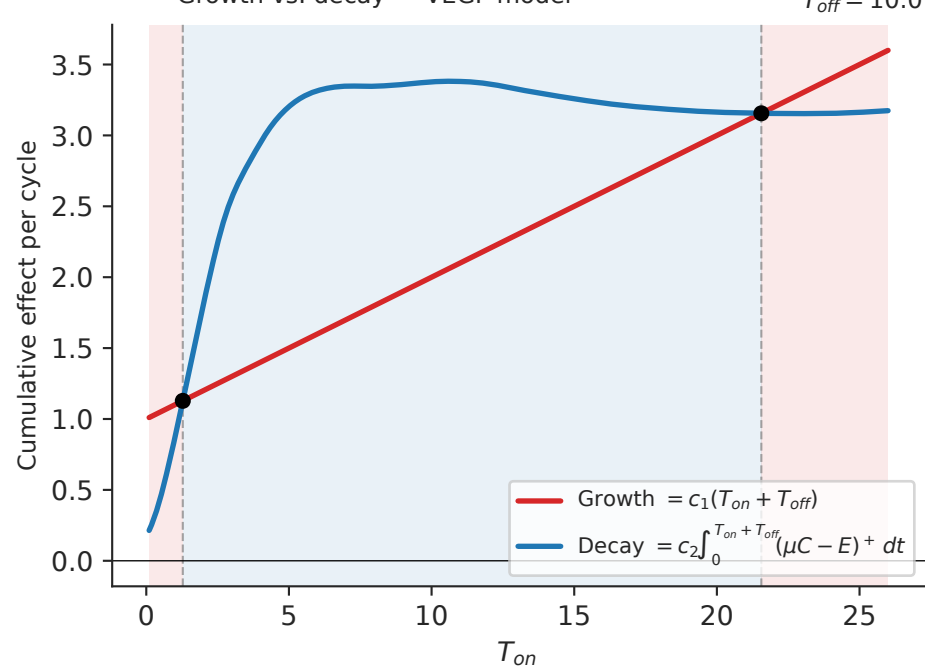
$$\frac{dE}{dt} = E \left[(a_1 A_1 + a_2 A_2) \left(1 - \frac{E}{4\mu C_0} \right) - r_E \right]$$

$$\frac{dC}{dt} = C(c_1 - c_2(\mu C - E)^+)$$

A_1 : VEGF, A_2 : compensatory signal
 E : vasculature, C : tumor size

**F**

Growth vs. decay — VEGF model

 $T_{off} = 10.0$ **G**

Adaptive therapy for the optimal treatment time found
 $(T_{on} = 5.6, T_{off} = 10.0)$

